

RANDOM VARIABLE

A variable is random if it takes on different values as result of the outcomes of a random experiment or random event. A random variable assigns a numerical value to each outcome of a random experiment and the type depends on the set of values it can take.

A random variable is **discrete** if it has a finite or countable number of possible outcomes often represented by whole numbers, other than this they can also be represented by other discrete values.
Ex: No. of heads in 10 coin flips.

• A random variable is **Continuous** if it has an uncountable number of possible outcomes within a given interval (a, b) . That means that each and every single point in the interval can be taken up as the outcome values when you do the experiment. Hence, you do not have discrete values in this set of possible values but rather an interval.

Thus, $X = \{x: x \text{ belongs to } (a, b)\}$

Example of a Continuous Random Variable

- The height of an adult male or female. $(145 < x < 170)$
- The speed of a vehicle on a highway $(80 < x < 120)$
- The temperature of a patient $(96 < x < 106)$

The key differences between discrete and continuous random variables

	Discrete Random Variable	Continuous Random Variable
Definition	Takes on a finite or countably infinite set of possible values.	Takes on any value within a range or interval
Probability Distribution	Described by a probability mass function (PMF), which gives the probability that a discrete variable takes for an exact value: $P(X=x)$. The sum of PMF for all possible values equals 1, and each value is non negative.	Described by a probability density function (PDF), which gives the probability density at each possible value, not for exact points but for an interval. Probabilities are assigned to interval and PDF is used to compute these probabilities by integrating over the interval. The total area under the PDF curve is 1

	Discrete Random Variable	Continuous Random Variable
Example	Number of heads in three coin tosses.	Height of a person selected at random.
Probability of a single value	Probability at each possible value is non zero	It is zero probability at each possible value.
Cumulative Distribution Function	Describes the probability of getting a value less than or equal to a particular value.	Describes the probability of getting a value less than or equal to a particular value.
Mean and Variance	Mean and variance can be calculated directly from the PMF.	Mean and variance can be calculated using the PDF and integration.

Suppose you want to know the probability that either (a) you get tails and roll a “6” or (b) you get heads and the person’s height is between 160 and 180 cm.

- $P(\text{Tails and Die} = 6) = P(\text{Tails}) \times P(\text{Die} = 6) = 0.5 \times 1/6 = 1/12$
- $P(\text{Heads and } 160 \leq \text{height} \leq 180) = 0.5 \times P(160 \leq X \leq 180)$
Here, $P(160 \leq X \leq 180)$ is calculated by integrating the normal PDF from 160 to 180.

You would add both contributions for the final probability since the events are mutually exclusive.

CONCEPT OF CORRELATION AND REGRESSION

Correlation measures how strongly two variables move together, while regression models the relationship to predict one from the other. Both are key tools in statistics for analyzing data patterns.

Correlation Basics:

Correlation is the statistical tool used to analyze and measure the degree of relationship between two and more than two variables, as how change in the values of one variable are associated with changes in the values of other variable.

Correlation does not imply **causation** (one event directly causes another event to happen); it only describes association.

Correlation quantifies the strength and direction of the linear association between two variables, The degree of relationship is measured using **Pearson's correlation coefficient (r)** ranging from **-1 to +1**. **A value near 1 shows a strong positive linear relationship** (where both variables increases or decreases together), and **a value near -1 indicates a strong negative linear relationship** (where one variable increases and other decreases). **A value 0 means no linear relationship**. It treats both variables symmetrically, so $\text{correlation}(X,Y)$ equals $\text{correlation}(Y,X)$ and this doesn't imply causation.

+1 indicates perfect positive correlation

-1 indicates perfect negative correlation

Correlation is unit-free (standardized), so you can compare correlations across different scales.

Ex: If **study hours increase and marks increase**, there is **positive correlation**.

If **price increases and demand decreases**, there is **negative correlation**.

If we have variables X = advertising expenditure and Y = sales then, if higher advertising leads to higher sales then variables are positively correlated.

The problem of analyzing relationship between variables can be divided into following steps—

- Determine if the relationship exists between the variables. (We can estimate value of one variable given value of other variable)
- If it exists, then measure it using measure of correlation
- Testing whether it is significant
- Establish the cause and effect relation if any---→ regression analysis

Correlation Analysis helps in understanding the economic behavior, estimating cost, price, sales and other variables.

Regression Basics:

Regression describes how an independent variable (X) predicts a dependent variable (Y), typically via a line equation like $Y = a + bX$, where **b is the slope** showing change in Y per unit change in X . Unlike correlation, it distinguishes predictor from outcome, and swapping variables yields different equations.

The problem of analyzing relationship between variables can be divided into following steps—

Determine if the relationship exists between the variables. (We can estimate value of one variable for given value of other variable). If it exists, then measure it using **measure of correlation**.

Testing if the measure is significant, find the **covariance** to shows how variables vary together and then establish the **cause and effect** relation if any---→ **regression analysis**.

CONCEPT OF CORRELATION

Types of correlation:

On the basis of the nature of the relationship correlation may be :

- (1) **Positive and Negative Correlation (based on direction)**
- (2) **Linear and Curvilinear Correlation (based on pattern)**
- (3) **Simple Multiple and Partial Correlation (based on no.of variables)**

Positive and Negative Correlation:

When the two variables change in the same direction, that is, both increase or both decrease the relationship is said to be **positive or direct**.

But when the change is in opposite direction that is, one is increasing the other is decreasing the correlation is **negative or inverse**.

In Positive Correlation variables change in same direction ($0 < r \leq 1$)

As X increases, Y increases;

As X decreases, Y decreases

X	10	12	15	18	20	24	30
Y	15	20	25	30	35	40	45

In Negative Correlation variables change in opposite direction. ($-1 \leq r < 0$)

As X increases, Y decreases;

As X decreases, Y increases

X	10	12	15	18	20	24	30
Y	45	40	35	30	25	20	15

In Zero Correlation there is no relationship between variables ($r = 0$)

Ex: relation between shoe size and intelligence.

Examples : (Positive Correlations)

(Negative Correlation)

- 1) Age of husband and age of wife
- 2) Income of a consumer and expenditure

- Price of commodity and quantity demand
- Expenditure and savings

Linear and Curvilinear Correlation:

If the ratio of change between two variables is uniform then there exist linear correlation between them. The linear relationship is described by a **straight line**.

In case of non-linear relationship, the amount of change in one variable does not bear a constant ratio to the amount of change in the other variable. Such relationship is described by a **curve**.

Simple, Multiple and Partial Correlation:

Simple, multiple and partial correlation is based upon the **number of variables** under study. When only **two variables** are studied, the analysis of the relationship between them is called **simple correlation**.

When **three or more variables** are studied, the relationship can be either **multiple or partial**.

In multiple correlation, three or more variables are studied simultaneously. But in partial correlation we consider only two variables influencing each other while the effect of other variable(s) is held constant.

Methods of Studying Correlation

Methods of studying correlation primarily include

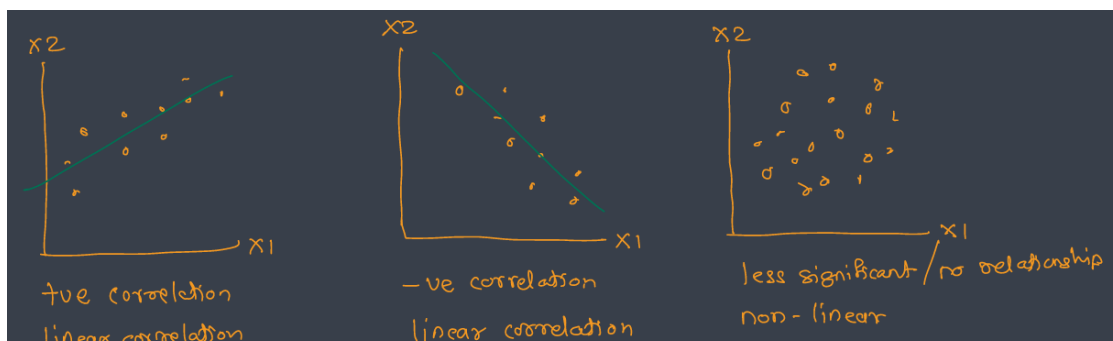
- Scatter Diagram Method
- Karl Pearson's Coefficient of Correlation
- Spearman's Rank Correlation Coefficient.

1.Scatter Diagram

The simplest way of asserting whether two variables X and Y are related or not is to prepare a dot chart, to observe the pattern and direction of their relationship. This dot chart is called as **Scatter Diagram**. Each point represents the values of X & Y as a coordinate.

It is simple, graphical and non- mathematical method not influenced by the extreme values. But **it cannot adopt the exact degree of correlation.**

For example, plotting heights and weights of individuals to see if taller people tend to weigh more.



Interpretation:

- Dots going upward as you move right → **positive correlation**.
- Dots going downward as you move right → **negative correlation**.
- Dots forming a narrow band (close to a straight line) → **strong correlation**.
- Dots widely scattered with no pattern → **little or no correlation**.

Advantages:

- Very simple and visual.
- Good for detecting form of relationship (linear, curved, clusters, outliers).

Limitations:

- Does **not** give a numerical value of correlation.
- Judgment can be subjective (“strong” vs “moderate”).

2. Karl Pearson's Coefficient of Correlation

A mathematical method for measuring the magnitude of linear relationship between two continuous variables was suggested by Karl Pearson (1867-1936), a great British Statistician.

Karl Pearson's Coefficient of Correlation between two variables X and Y is denoted by $r(X,Y)$ or r_{xy} . The coefficient of correlation ‘r’ measures the degree of strength and direction of linear relationship between two continuous variables say x & y and is calculated by comparing **covariances of variables normalized by their standard deviations** [(ratio of the (covariance between X and Y) to (the product of the standard deviations of X and Y))]

$$r = \frac{\Sigma(X-\bar{X})(Y-\bar{Y})}{\sqrt{\Sigma(X-\bar{X})^2} \sqrt{\Sigma(Y-\bar{Y})^2}} \quad \begin{array}{l} \bar{X} = \text{mean of X variable} \\ \bar{Y} = \text{mean of Y variable} \end{array}$$

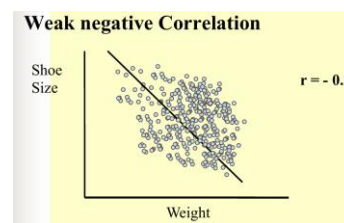
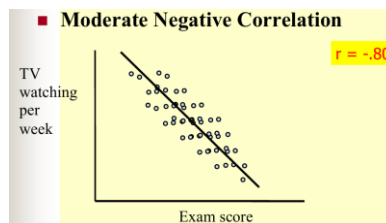
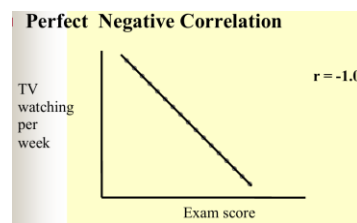
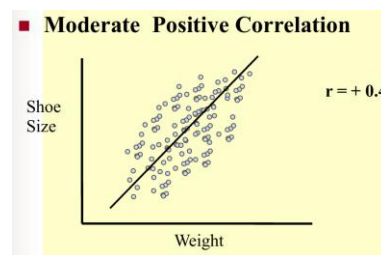
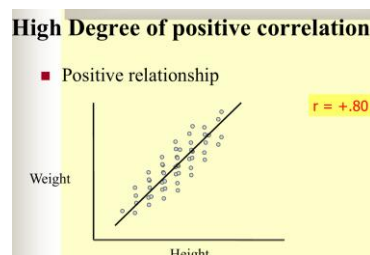
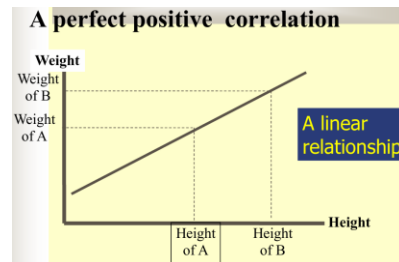
$$r = \text{Cov}(XY) / \text{SD}_x * \text{SD}_y$$

The Karl Pearson's coefficient ‘r’ ranges from -1 to +1 and is unit-free.

Degree of Correlation

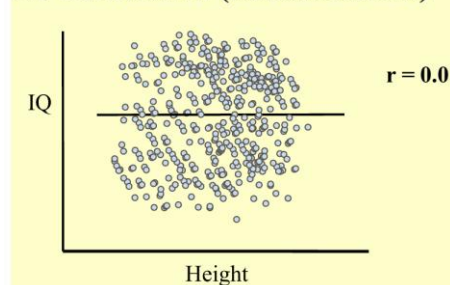
The relationship between two variables can be determined by the quantitative value of coefficient of correlation (r).

Perfect Correlation: When changes in two variables are exactly proportional, correlation is said to be perfect. If equal proportional changes are in the same direction then there is **perfect positive correlation** and is describe as +1; and if equal proportional changes are in opposite direction ,there is **perfect negative correlation** and is described as -1.



Zero correlation : If the two variables are uncorrelated then there is zero correlation.

No Correlation (horizontal line)



Limited Degree of correlation :

Correlation is said to be **limited positive** when there is unequal changes in the two variables in the same direction and correlation is **limited negative** when there are unequal changes in the opposite direction.

The limited degree of correlation can be high(between ± 0.75 to 1),moderate (± 0.25 to 0.75) or low (between ± 0 to 0.25).

Degree of Correlation	Positive	Negative
Perfect Correlation	+1	-1
Very high degree of Correlation	+0.9	-0.9
Moderate degree of Correlation	+.75 to +0.9	-.75 to -.9
Only the possibility of correlation	+.5 to+.75	-.5 to -.75
Low Correlation	+.3to +.5	-.3 to-.5
Zero Correlation	0	0

Correlation Coefficient can be calculated without taking deviations of the values either from actual mean or assumed mean

$$r = \frac{N \sum XY - \sum X \sum Y}{\sqrt{N \sum X^2 - (\sum X)^2} \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

Degree of Correlation is expressed by a value of Coefficient

Direction of change is Indicated by sign (- ve) or (+ ve)

Interpretation of Correlation Coefficient (r)

- The value of correlation coefficient 'r' ranges from -1 to +1. (- 1 ≤ r ≤ 1)
- If r = +1, then the correlation between the two variables is said to be perfect and positive .
- If r = -1, then the correlation between the two variables is said to be perfect and negative.
If r = 0, then there exists no correlation between the variables, the variables are not related.
- The correlation coefficient is independent of the change of origin & scale.
- The coefficient of correlation is the geometric mean of two regression coefficient
 $r = \sqrt{b_{xy} * b_{yx}}$
- The one regression coefficient is (+ve) other regression coefficient is also (+ve)
correlation coefficient is (+ve)

Karl Pearson's coefficient summarizes in one value, the degree of correlation & direction of correlation also.

Limitation of Pearson's Coefficient

- Always assume linear relationship
- Interpreting the value of r is difficult.
- Value of Correlation Coefficient is affected by the extreme values.
- Time consuming methods

Coefficient of Determination

The convenient way of interpreting the value of correlation coefficient is to use of **square of coefficient of correlation** which is called **Coefficient of Determination**.

The Coefficient of Determination = r^2 . Suppose: $r = 0.9$, $r^2 = 0.81$ this would mean that 81% of the variation in the dependent variable has been explained by the independent variable.

The maximum value of r^2 is 1 because it is possible to explain all of the variation in y but it is not possible to explain more than all of it.

Coefficient of Determination = Explained variation / Total variation

Example:

Suppose: $r = 0.60$ and $r = 0.30$. It does not mean that the first correlation is twice as strong as the second the 'r' can be understood by computing the value of r^2 .

When $r = 0.60$, $r^2 = 0.36$ -----(1)

$r = 0.30$, $r^2 = 0.09$ -----(2)

This implies that in the first case 36% of the total variation is explained whereas in second case 9% of the total variation is explained.

3. Spearman's Rank Coefficient of Correlation

This method measures the strength and direction of the monotonic relationship between two variables (rate of increase or decrease is not constant). It is used when the data is ordinal (can be arranged in serial order) or when the assumptions for Pearson's correlation (like linearity and normality) are not met. To use it, **data points are ranked**, and the differences in ranks are used to calculate the correlation coefficient **R** which also ranges between -1 and +1.

$$R = 1 - (6 \sum D^2) / N (N^2 - 1)$$

R = Rank correlation coefficient

D = Difference of rank between paired item in two series.

N = Total number of observation

Example: A teacher wants to study the relationship between the marks in statistics and in python of 8 students

Student	Statistics (X)	Python (Y)
A	85	80
B	78	75
C	92	95
D	70	65
E	60	70
F	88	85
G	75	78
H	68	60

Assign ranks. The highest value gets highest rank

Student	Rank X	Rank Y	d = Rx - Ry	d ²
A	3	3	0	0
B	4	5	-1	1
C	1	1	0	0
D	6	7	-1	1
E	8	6	2	4
F	2	2	0	0
G	5	4	1	1
H	7	8	-1	1

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$\begin{aligned}
 r_s &= 1 - \frac{6(8)}{8(8^2 - 1)} \\
 &= 1 - \frac{48}{8(63)} \\
 &= 1 - \frac{48}{504} \\
 &= 1 - 0.0952
 \end{aligned}$$

Example 2:

Two judges rank 5 students in a competition:

Student	Rank by Judge 1 (X)	Rank by Judge 2 (Y)
A	1	2
B	2	1
C	3	4
D	4	3
E	5	5

Student	X	Y	d = X-Y	d ²
A	1	2	-1	1
B	2	1	1	1
C	3	4	-1	1
D	4	3	1	1
E	5	5	0	0

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \qquad \sum d^2 = 1 + 1 + 1 + 1 + 0 = 4$$

$$r_s = 1 - \frac{6 \times 4}{5(5^2 - 1)}$$

$$r_s = 1 - \frac{24}{5(25 - 1)}$$

$$r_s = 1 - \frac{24}{5 \times 24}$$

$$r_s = 1 - \frac{24}{120}$$

$$r_s = 1 - 0.2 = 0.8$$

$r_s = 0.8$

which indicates strong positive correlation.

The judges have similar rankings.

Example 3:

Calculate spearman rank correlation for scores of 5 students in maths and physics

Student	Math Score (X)	Physics Score (Y)
1	85	92
2	70	65
3	90	80
4	75	78
5	55	60

Student	X Score	RX (Rank of X)	Y Score	RY (Rank of Y)
1	85	2	92	1
2	70	4	65	4
3	90	1	80	2
4	75	3	78	3
5	55	5	60	5

Student	RX	RY	$d_i = RX - RY$	d_i^2
1	2	1	$2 - 1 = 1$	1
2	4	4	$4 - 4 = 0$	0
3	1	2	$1 - 2 = -1$	1
4	3	3	$3 - 3 = 0$	0
5	5	5	$5 - 5 = 0$	0

$$\sum d_i^2 = 1 + 0 + 1 + 0 + 0 = 2$$

$$n = 5$$

$$r_s = 1 - \frac{6 \times 4}{5(5^2 - 1)}$$

$$r_s = 1 - \frac{24}{5(25 - 1)}$$

$$r_s = 1 - \frac{24}{5 \times 24}$$

$$r_s = 1 - \frac{24}{120}$$

$$r_s = 1 - 0.2 = 0.8$$

Interpretation of Rank Correlation Coefficient (R)

- The value of rank correlation coefficient, R ranges from -1 to +1
- If R = +1, then there is complete agreement in the order of the ranks and the ranks are in the same direction
- If R = -1, then there is complete agreement in the order of the ranks and the ranks are in the opposite direction
- If R = 0, then there is no correlation

If ranks are not given, it is needed to assign ranks to the data series.

Merits and Limitations of Spearman's Correlation

- It can detect monotonic association, but if the relationship is non-monotonic (the curve goes up and down), this method fails.
- This method is simpler to understand and easier to apply than Pearson's coefficient.
- The method is useful where we can give ranks and not the actual data.
- It cannot be used in finding correlation for grouped frequency distribution.
- The method should be applied when N exceeds 30.

REGRESSION ANALYSIS

Regression Analysis is a very powerful tool in the field of statistical analysis in predicting the value of one dependent variable, given the value of another independent variable, when those variables are related to each other.

Regression helps determine the **strength and nature** of associations between variables, **predict future outcomes** based on past data, and **analyze the effects** of one or more factors on a variable of interest. It is widely applied in economics, finance, healthcare, real estate, marketing, data science, and numerous scientific fields.

Advantages of Regression Analysis

- Linear regression is widely used because it is simple, easy to interpret, and computationally efficient.
- It quantifies the relationship between variables, helps make predictions, forecasting and provides diagnostic tools (like residual analysis) to check model quality.
- It's also a foundational method for more advanced regression techniques.

REGRESSION LINE

Regression line is the line which gives the best estimate of one variable from the value of any other given variable.

The regression line gives the average relationship between the two variables in mathematical form.

For two variables X and Y, there are always two lines of regression

Regression line of X on Y: gives the best estimate for the value of X for any specific given values of Y

$X = a + b Y$ where

- a = x-intercept (value of x when $y=0$)
- b = Slope of the line (rate of change of x with respect to y)
- X = Dependent variable
- Y = Independent variable

It is chosen so that the **sum of squared vertical distances** between the points and the line is as small as possible (least squares idea).

For the regression equation $x = a + by$, the intercept a as a fraction in terms of n and sums can be written as:

$$a = \frac{\sum x \sum y^2 - \sum y \sum xy}{n \sum y^2 - (\sum y)^2}$$

Similarly, the slope b is:

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

Regression line of Y on X : gives the best estimate for the value of Y for any specific given values of X

$Y = a + bX$ where

- X = independent variable
- Y = Dependent variable
- a is the y-intercept (value of y when x = 0)
- b is the slope of the line (rate of change of y with respect to x)

The coefficients a and b are calculated using the formulas:

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$$

Here, n is the number of data points, $\sum xy$ is the sum of the products of paired scores, $\sum x$ and $\sum y$ are sums of the x and y values respectively, and $\sum x^2$ is the sum of squared x values.

Interpretation of Regression Lines

- In case of perfect correlation (positive or negative) the two line of regression coincide.
- If the two R. line are far from each other then degree of correlation is less, & vice versa.
- The mean values of X & Y can be obtained as the point of intersection of the two regression line.
- The higher degree of correlation between the variables, the angle between the lines is smaller & vice versa.

Regression Lines when deviations are taken from Mean

The regression equation or line when deviations are taken from the arithmetic mean (also called the centered form) is:

$$(y - \bar{y}) = b(x - \bar{x})$$

Where:

- $\bar{x}$ and $\bar{y}$ are the means of the x and y variables respectively,
- b is the regression coefficient (slope),
- $(x - \bar{x})$ and $(y - \bar{y})$ are the deviations of x and y from their means.

In this form, the regression line always passes through the means $(\bar{x}, \bar{y})$, and the intercept equals zero, simplifying interpretation.

The slope b is calculated as:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\mathbf{b_{yx} = r (\sigma_y / \sigma_x)}$$

The slope b in the regression equation expressed as $(x - \bar{x}) = b(y - \bar{y})$, where variables are centered by subtracting their means, is given by:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(y_i - \bar{y})^2}$$

This is the ratio of the covariance between x and y to the variance of y . In other words, it measures how much x changes, on average, for a unit change in y about their means.

$$\mathbf{b_{xy} = r (\sigma_x / \sigma_y)}$$

Regression Coefficients and Correlation

The slopes b_{yx} and b_{xy} are called **regression coefficients**.

For sample data:

- Regression of y on x:

$$b_{yx} = \frac{\text{Cov}(X, Y)}{\sigma_X^2}$$

- Regression of x on y:

$$b_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_Y^2}$$

They are related to the correlation coefficient r by:

$$r^2 = b_{yx} \times b_{xy}$$

If b_{xy} is positive then b_{yx} should be positive and vice-versa.

If one regression coefficient is greater than one, the other must be less than it.

Arithmetic mean of regression coefficients is equal to or greater than coefficient of correlation

$$b_{yx} + b_{xy} / 2 \geq r$$

The sign of r is same as the sign of each regression coefficient. So,

- if correlation is positive, both regression lines slope upwards.
- If correlation is negative, both slope downwards.
- If correlation is zero, the regression lines become horizontal/ vertical (no useful prediction)

Correlation tells **how strong** the linear association is; regression tells **how to predict** one variable from the other.

Standard Error of Estimate: It measures the average distance that the observed values fall from the regression line. It quantifies the accuracy of predictions made by the regression model.

$$SEE = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

Where y_i are observed values ; $\hat{y}_i$ are predicted values from regression line,

n = no of observations ; $n-2$ represents degree of freedom (slope and intercept are estimated)

A smaller SEE indicates a better fit, meaning the predictions are closer to actual values. It has the same units as the dependent variable y .

Standard error of estimate (SEE) of Y measure the variability of the observed values of Y around the regression line.

The standard error of the estimate (SEE) in terms of the regression intercept a and slope b can be expressed as:

$$SEE = \sqrt{\frac{\sum (y_i - a - bx_i)^2}{n - 2}}$$

where:

- y_i are the observed values,
- x_i are the independent variable values,
- a is the intercept,
- b is the slope,
- n is the number of observations.

Or

$$(SE_{xy}) = \sqrt{\frac{\sum Y^2 - a \sum Y - b \sum YX}{n - 2}}$$

for line $Y = a + bX$ where

X = value of independent variable

Y = value of dependent variable

a = y-intercept

b = slope

n = no. of observations

Correlation analysis vs. Regression analysis.

- a) Regression is the average relationship between two variables
- b) Correlation need not imply cause & effect relationship between the variables understudy. Regression Analysis clearly indicate the cause and effect relationship between the variables.
- c) There may be non-sense correlation between two variables.- There is no such thing like non-sense regression
- d) Regression is the average relationship between two variables.

COVARIANCE

Covariance is a statistical measure that describes how two variables change together, the degree and direction to which they vary with respect to each other, but not about the strength of relationship.

It is an essential tool used in various fields, including finance, economics, and science.

Covariance is calculated by taking the **average of the product of the deviations** of each variable from their respective **means**.

Covariance can be positive (they tend to increase together), negative (one tends to increase when the other decreases), or zero (no linear relationship detected).

If student **study more** will tend to get **more marks** (positive covariance)

If student **play more** will tend to get **less marks** (negative covariance)

For two random variables X and Y ,

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Covariance Formula

For Population:

$$\text{Cov}_p(x, y) = \frac{\sum_{i=1}^n (x_i - x') (y_i - y')}{n}$$

For Sample:

$$\text{Cov}_s(x, y) = \frac{\sum_{i=1}^n (x_i - x') (y_i - y')}{n - 1}$$

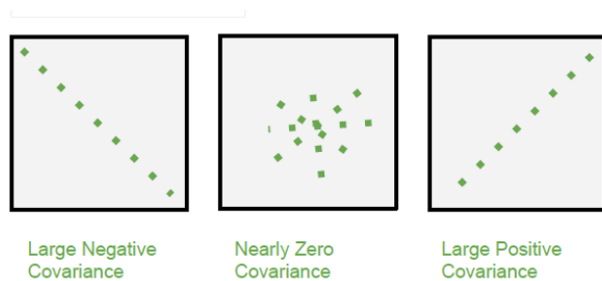
Here, x' and y' = mean of given sample set n (total no of sample x_i and y_i)

The population covariance uses the true population mean, while the sample covariance uses the sample mean, which is an estimate. Dividing by $n - 1$ instead of n corrects bias in the estimation from a sample.

It can take any value between $-\infty$ to $+\infty$ where the negative value represents the negative relationship whereas a positive value represents the positive relations.

Interpretation:

- $\text{Cov} > 0$: X and Y tend to move in the same direction. (**Positive Covariance**)
- $\text{Cov} < 0$: X and Y tend to move in opposite directions. (**Negative Covariance**)
- $\text{Cov} = 0$: no linear association, they move independent of each other (**Zero Covariance**).



Covariance is in units (unit of X)×(unit of Y). Its magnitude depends on variable scales, so it's hard to compare across datasets.

If X and Y are independent (for many common distributions), $\text{Cov}(X,Y) = 0$. The converse is not always true.

Example –

X (study hour)	2	4	6	8
Y(marks)	50	60	65	80

$$\text{Mean}(X) = (2 + 4+6+8)/4 = 5$$

$$\text{Mean}(Y) = (50+60+65+80)/4 = 63.75$$

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
2	50	-3	-13.75	41.25
4	60	-1	-3.75	3.75
6	65	1	1.25	1.25
8	80	3	16.25	48.75

$$\sum (X - \bar{X})(Y - \bar{Y}) = 95 \qquad \text{Cov}(X, Y) = \frac{95}{3} = 31.67$$

Covariance is positive, so study hours and marks increase together.

Properties of Covariance

- $\text{Cov}(x, x) = \text{Var}(x)$ [Covariance of a variable with itself equals its variance]
- $\text{Cov}(x, y) = \text{Cov}(y, x)$ [Covariance is symmetric]
- $\text{Cov}(x, \text{constant}) = 0$
- $\text{Cov}(x, y) = 0$ [If x, y are independent]

Covariance Matrix

When we have multiple variables, Covariance is represented using a matrix.

For variables x, y, z, the **Covariance matrix** is

$$\begin{bmatrix} Var(X) & Cov(X,Y) & Cov(X,Z) \\ Cov(Y,X) & Var(Y) & Cov(Y,Z) \\ Cov(Z,X) & Cov(Z,Y) & Var(Z) \end{bmatrix}$$

It is widely used in Machine Learning, Principal Component Analysis (PCA).

Relation between Correlation(r) and Covariance is

The numeric value of Covariance depends on the units of X and Y, so it's hard to interpret magnitude directly. That's why we prefer correlation, which is the normalized version of covariance:

$$\text{Correlation}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where σ_X and σ_Y are standard deviations of X and Y.

Covariance	Correlation
Covariance is a measure of how much two random variables vary together	Correlation is a statistical measure that indicates how strongly two variables are related.
Involves the relationship between two variables or data sets	Involves the relationship between multiple variables as well
Lie between -infinity and +infinity	Lie between -1 and +1
Measure of correlation	Scaled version of covariance
Provides direction of relationship	Provides direction and strength of relationship
Dependent on scale of variable	Independent on scale of variable
Have dimensions	Dimensionless

8. Covariance vs Variance

Feature	Variance	Covariance
Variables	ONE variable	TWO variables
Formula	$\Sigma(x_i - \bar{x})^2 / n$	$\Sigma(x_i - \bar{x})(y_i - \bar{y}) / n$
Measures	Spread of one variable	Joint variation of two
Result	Always ≥ 0	Can be +, -, or 0
Special case	—	$\text{Cov}(X, X) = \text{Var}(X)$
Units	Units ²	Units of X $\times$ Units of Y

OUTLIERS

An outlier is an observation that lies at an abnormal distance from other values in a random sample from a population. It is unusually high, low or different from the general pattern.

```
Data:  2  3  3  4  4  5  5  6  6          47  ← Outlier!
      |___ Normal cluster ___|      |
                        Far away
```

Mean, Standard Deviation, Correlation and range are highly affected by outliers.

Always visualize your data and check for outliers before computing Covariance or Correlation, since a single extreme point can completely change your conclusions.

How to Detect Outliers

Method 1 — IQR (Interquartile Range) Rule

1. Find Q1 (25th percentile) and Q3 (75th percentile)
2. $\text{IQR} = Q3 - Q1$
3. Any point **below** $Q1 - 1.5 \times \text{IQR}$ or **above** $Q3 + 1.5 \times \text{IQR}$ is an outlier

Method 2 — Z-Score Rule

$$Z = (X - \mu) / \sigma$$

If $|Z| > 3$, the point is more than 3 standard deviations from the mean → **likely an outlier**

Method 3 — Visual Methods:

- **Box plots** — points beyond whiskers are outliers
- **Scatter plots** — visually obvious isolated points
- **Histograms** — lone bars far from the main distribution